University of Ottawa

Department of Mathematics and Statistics

MAT 2143B : Introduction to Group Theory

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Midterm Exam

March 1, 2023

Surname _____ First Name _____

Student # _____

Instructions:

(a) You have 80 minutes to complete this exam and submit your PDF file on BrightSpace.
(b) All work to be considered for grading should be written on your solution sheets. You should clearly indicate where the solution of each question is written.
(c) No notes, books, scrap paper, calculators or other electronic devices are allowed.
(d) You are strongly recommended to write in **pen**, not pencil.

Please do not write in the table below.

Question	1	2	3	4	5	6	7	Total
Maximum	6	6	4	5	4	4	2	31
Grade								

1. (a) **[3 points]** Determine all of the elements $x \in U(211)$ with the property that $x = x^{-1}$, where as usual x^{-1} means the inverse of x in the group $U(211)$.

Hint: 211 is a prime number.

(b) **[3 points]** Determine all of the elements $x \in U(633)$ with the property that $x = x^{-1}$.

Solution: (a) $x = x^{-1}$ is equivalent to $x^2 = 1$. We have $211|x^2 - 1$ if and only if $211|(x-1)(x+1)$. Since 211 is prime, the latter is equivalent to

$$211|x-1 \quad \text{or} \quad 211|x+1.$$

When $211|x-1$ we obtain $x = 1$ (because $0 \leq x \leq 211$) and when $211|x+1$ we obtain $x = 210$.

Final answer: $x = 1$ or $x = 210$.

(b) As in (a), we have $633|(x-1)(x+1)$. There are the following cases:

Case I. $633|x-1 \Rightarrow x = 1$ (recall that $0 \leq x \leq 632$).

Case II. $633|x+1 \Rightarrow x = 632$ (recall that $0 \leq x \leq 632$).

Case III. $3|x-1$ and $211|x+1$. Since $0 \leq x \leq 632$, from the second relation we have $x+1 \in \{0, 211, 422\}$, hence $x \in \{632, 210, 421\}$. Since we must also have $3|x-1$, by checking the values it follows that the only acceptable value is $x = 421$.

Case IV. $3|(x+1)$ and $211|(x-1)$. The analysis is similar to Case III. In this case $x = 212$.

Final answer: $x \in \{1, 212, 421, 632\}$

2. (a) **[2 points]** Write down the definition of a group.

Solution: This is Def. 2.15 from the notes.

A *group* is a set G equipped with a binary operation $\star : G \times G \to G$ which satisfies the following:

- The binary operation $\star$ is *associative*, i.e., $(a \star b) \star c = a \star (b \star c)$ for all $a, b, c \in G$.
- G has an *identity element*. That is, there exists an element $e \in G$ such that $a \star e = e \star a = a$ for all $a \in G$. Sometimes e is called the *neutral element* of G.
- Every $a \in G$ has an *inverse*, i.e., for every $a \in G$ there exists an element $a^{-1} \in G$ such that $a \star a^{-1} = a^{-1} \star a = e$.

(b) **[4 points]** Let $G := \mathbb{Z} \times \mathbb{Z} \times \mathbb{Z}$ and let $H := \{(x, y, z) \in \mathbb{Z} \times \mathbb{Z} \times \mathbb{Z} : x + 2y = -3z\}$. Using the Concise Subgroup Test, prove that H is a subgroup of G.

Solution: Given $(a, b, c) \in H$ and $(a', b, c') \in H$ we need to show that

$$(a, b, c) - (a', b', c') = (a - a', b - b', c - c') \in H.$$

We have $a + 2b = -3c$ and $a' + 2b' = -3c'$. Thus

$$(a - a') + 2(b - b') = a + 2a' - (b + 2b') = -3c - (-3c') = -3(c - c').$$

This proves $(a-a', b-b', c-c') \in H$. Since H is also nonempty (for examples $(0, 0, 0) \in H$) it follows that $H \leq G$.

3. [**4 points**] Let $G := \mathbb{Z}_{36} \times \mathbb{Z}_3$. Determine all of the elements $g \in G$ such that $o_G(g) = 4$. You should justify your answer.

Solution: Let $g = (a, b)$ with $a \in \mathbb{Z}_{36}$ and $b \in \mathbb{Z}_4$. Then

$$(0, 0) = g^4 = (4a, 4b) \;\Rightarrow\; 4a = 0 \text{ in } \mathbb{Z}_{36} \text{ and } 4b = 0 \text{ in } \mathbb{Z}_3.$$

Thus $36|4a$ and $3|4b$. Since 3 is prime and $\gcd(3, 4) = 1$ it follows that $3|b$ hence $b = 0$ in $\mathbb{Z}_3$. Next note that

$$36|4a \;\Rightarrow\; 4a = 36k,\; k \in \mathbb{Z} \;\Rightarrow\; a = 9k,\; k \in \mathbb{Z}.$$

It follows that $a \in \{0, 9, 18, 27\}$. So far we obtain

$$(a, b) \in \{(0, 0), (9, 0), (18, 0), (27, 0)\}$$

By checking directly, we see that the elements of order 4 in the above list are

$$(9, 0) \text{ and } (27, 0).$$

4. (a) **[2 points]** Write down the definition of the centre of a group G.

Solution: The centre of G is the set of elements $g \in G$ such that $gh = hg$ for all $h \in G$.

(b) **[2 points]** Write down the proof of the following assertion: the centre of a group G is always a subgroup of G.

Solution: We use the concise subgroup test. For $a, b \in Z(G)$ we have $ax = xa$ and $bx = xb$ for all $x \in G$. The second relation after left and right multiplication by b^{-1} yields $b^{-1}x = xb^{-1}$. Thus for every $x \in G$ we have

$$ab^{-1}x = a(b^{-1}x) = axb^{-1} = (ax)b^{-1} = xab^{-1},$$

that is, $ab^{-1} \in Z(G)$.

(c) **[1 point]** Give an example of a group G such that $Z(G) \neq \{e\}$ and $Z(G) \neq G$. You should justify your answer.

Solution: There are many examples. For instance $G = Q = \{\pm 1, \pm i, \pm j, \pm k\}$ whose centre is $\{\pm 1\}$ (this follows from the relations $ij = -ji$, $ik = -ki$, $kj = -jk$). Another example is any group of the form $G \times H$ where G is non-abelian and H is abelian. Then elements of the form (e, h) are in $Z(G \times H)$ but $Z(G \times H) \neq G \times H$ because $G \times H$ is non-abelian.

5. Let $\sigma \in S_8$ be the permutation

$$\sigma := (1\ 7\ 2\ 5\ 4)(3\ 8).$$

(a) **[3 points]** Compute the decomposition of σ^2 into a product of disjoint cycles. You should show your work.

Solution: We have $\sigma = \tau_1\tau_2$ where $\tau_1 = (1\ 7\ 2\ 5\ 4)$ and $\tau_2 = (3\ 8)$. Since τ_1 and τ_2 are disjoint we have

$$\sigma^2 = \tau_1^2\tau_2^2 = (1\ 2\ 4\ 7\ 5)\varepsilon = (1\ 2\ 4\ 7\ 5)$$

and this is the cycle decomposition of σ^2.

(b) **[1 point]** With σ as in (a), does the equation $\tau\sigma\tau = \sigma^2$ have a solution $\tau \in S_8$? You should justify your answer.

Solution: Since τ_1 is even and τ_2 is odd, the permutation σ is also odd. It follows that $\tau\sigma\tau$ is also an odd permutation: if τ is even then $\tau\sigma\tau$ is a product of two even and one odd permutations, hence it is odd. If τ is odd then $\tau\sigma\tau$ is a product of three odd permutations, so it is again odd.

However, σ^2 is an even permutation (since it is a 5-cycle, hence can be expressed as a product of 4 transpositions). Tus the left hand side of the equation is always an odd permutation, whereas the right hand side is an even permutation. This means that there does NOT exist any solution to this equaiton.

6. Let D_n denote the group of symmetries of a regular n-gon. Let $G := D_{12}$ and suppose that $r, s \in G$ denote the usual generators that satisfy $o_G(r) = 12$ and $o_G(s) = 2$.

(a) [**2 points**] How many elements does the cyclic subgroup of G generated by r^2 have? You should justify your answer.

Solution: THe order of r^2 is $12/\gcd(2, 12) = 12/2 = 6$. Hence by Theorem 6.6(iii) we have $\langle r^2 \rangle = \{e, r^2, r^4, r^6, r^8, r^{10}\}$, that is, it has 6 elements.

(b) [**2 points**] Simplify the expression rsr^4srs in G and write it as a product of the form $r^k s^\ell$ where $0 \leq k \leq 11$ and $0 \leq \ell \leq 1$.

Solution: We have $srs = r^{-1} = r^{11}$ and hence $sr = r^{11}s$. Thus

$$\begin{aligned} rsr^4(srs) &= rsr^4r^{-1} \\ &= rsr^4r^{11} \\ &= rsr^{15} \\ &= rsr^3 \\ &= rsrr^2 = rr^{11}sr^2 = rr^{11}(sr)r = rr^{22}sr = rr^{33}s = r^{34}s = r^{10}s, \end{aligned}$$

because $34 = 2 \times 12 + 10$, hence $r^{34} = r^{10}$.

7. [**2 points**] For each of the following statements, write **T** if the statement is true, and write **F** if the statement is false. You will receive .5 points for a correct answer, and -.5 points for an incorrect answer. (If you leave the answer space blank, you will not receive any negative points.) You do not need to justify your answers.

____ A d-cycle in S_n belongs to A_n if and only if d is even.

____ Every subgroup of $\mathbb{Z}$ other than the trivial subgroup is isomorphic to $\mathbb{Z}$.

____ The quaternion group $Q := \{\pm 1, \pm \mathbf{i}, \pm \mathbf{j}, \pm \mathbf{k}\}$ is not abelian, but it has an abelian subgroup with 4 elements.

____ If p is a prime number, then the cyclic group $\mathbb{Z}_{p^2}$ does not have any subgroups other than $\{0\}$ and $\mathbb{Z}_{p^2}$.

Solution: F, T, T, F.